

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks available |          |          |          | Maths    | Prac     |
|----------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Total    |          |          |
| 6        | (a) |      | No net force (1) Accept resultant of or sum of. Don't accept equal forces.<br>No net moment (1) Accept all or sum of CM = sum of ACM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2               |          |          | 2        |          |          |
|          | (b) |      | Moments taken about hinge. Clockwise moment shown to be $240 \times 2$ or 480 seen (1)<br>Anticlockwise moment: $T \sin 35^\circ \times 1.4$ seen (or $\cos 55^\circ$ ) (1)<br>$T = 597.8 \text{ [N]}$ (1)                                                                                                                                                                                                                                                                                                                                                                                                                            |                 | 3        |          | 3        | 3        |          |
|          | (c) | (i)  | 1.5 mm cable (1)<br>Safe working load > than 600 N [at 3:1 ratio] <b>or</b> only just above 600 N so to be safe will use 2 mm cable (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |          | 2        | 2        | 1        |          |
|          |     | (ii) | <b>Accept two</b> $\times$ (1) from the following options: <ul style="list-style-type: none"><li>• Shopkeeper does not have necessary expertise to decide on appropriate cable to use</li><li>• Should seek advice of engineer as suggested on web page or expert advice</li><li>• Shopkeeper should verify/ validate/ confirm/ authenticate wire rope data</li><li>• Taken rod to be weightless</li><li>• [If unsure] should have opted for higher diameter or applied higher safer working ratio</li><li>• Weather/high winds could affect tension</li><li>• Chosen wire does support the sign so made an informed choice</li></ul> |                 |          | 2        | 2        |          |          |
|          |     |      | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>2</b>        | <b>3</b> | <b>4</b> | <b>9</b> | <b>4</b> | <b>0</b> |